Inverse Variation and the Reciprocal Function

Name _____

Algebra 2 Lesson 4-1 (fast track) Date _____ Period _____

Learning goal: I can describe a translation of the reciprocal function, write its equation, and find its asymptotes, domain, and range.

Language goal: I can explain how a translation moves the graph of the reciprocal function using the words *translation*, *vertical asymptote*, *horizontal asymptote*, *domain*, and *range*.

Spanish: *asíntota* (asymptote), *función recíproca* (reciprocal function), *recíproco* (reciprocal).

On the quiz: (1) and (2) Inverse-variation tables (full Savvas quiz only). (3) Bonus, +5: a time that varies inversely with the number of students (Example 3, skipped here; the Extension practices it). (4) Graded, 10 points: the horizontal asymptote of a translated reciprocal function (Example 5). (5) Graded, 10 points: the equation of $y = \frac{1}{x}$ translated up and to the left (Example 5, then the Practice).

On the Topic 4 test: Question 5 asks you to describe the translation from $y = \frac{1}{x}$ to $y = \frac{1}{x+3} - 4$ (Example 5). Question 14 asks for every equation of $xy = 6$ translated up 3 and right 2 (Example 5). Question 1 (a width that varies inversely with a length) comes from Example 3, which this packet skips.

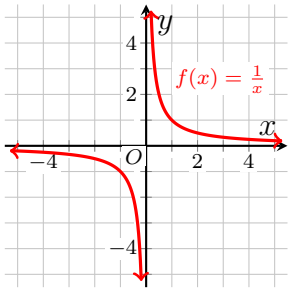
Example 5: Graph Translations of the Reciprocal Function

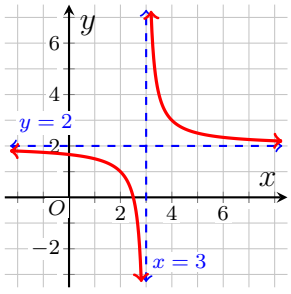
Graph $g(x) = \frac{1}{x-3} + 2$. What are the equations of the asymptotes? What are the domain and range?

Step 1 Use technology to graph the parent function, $f(x) = \frac{1}{x}$ (top graph). Its asymptotes are the axes, $x = 0$ and $y = 0$.

Step 2 Write g in terms of f: $g(x) = \frac{1}{x-3} + 2 = f(x-3) + 2$. So the graph of g is the graph of f **translated 3 units right and 2 units up** (bottom graph). The asymptotes move with it.

Step 3 The line $x = 3$ is a **vertical asymptote**. The line $y = 2$ is a **horizontal asymptote**. The **domain** is $\{x \mid x \neq 3\}$. The **range** is $\{y \mid y \neq 2\}$.





$g(x) = \frac{1}{x-3} + 2$

Key idea. Subtracting h from x translates the graph of f horizontally; adding k to $f(x)$ translates it vertically. So the graph of $g(x) = \frac{1}{x-h} + k$ is the graph of $f(x) = \frac{1}{x}$ translated h units horizontally and k units vertically. The asymptotes are $x = h$ and $y = k$; they cross at (h, k).

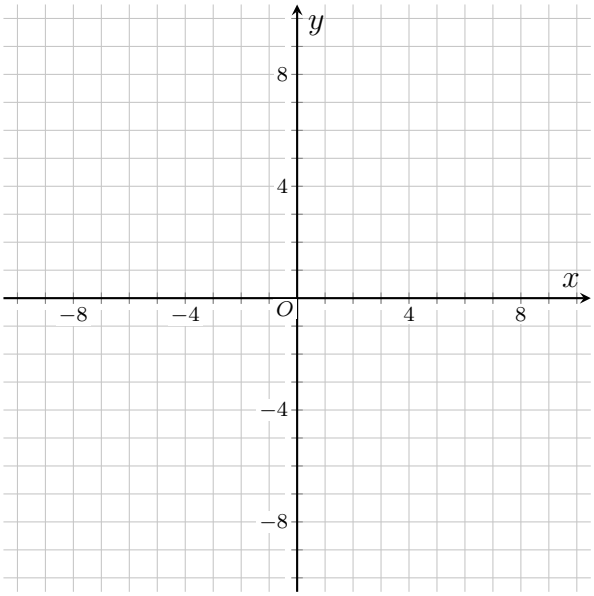
Concept Summary example: $y = \frac{1}{x-4} - 2$ has $h = 4$ and $k = -2$: right 4 and down 2; asymptotes $x = 4$ and $y = -2$.

Say it: The graph of g is a *translation* of $f(x) = \frac{1}{x}$: 3 units _____ and 2 units _____. The *vertical asymptote* is $x =$ _____, so the *domain* is every real number except _____. The *horizontal asymptote* is $y =$ _____, so the *range* _____.

Try It 5: Graph translations of the reciprocal function

Graph $g(x) = \dfrac{1}{x+2} - 4$. What are the equations of the asymptotes? What are the domain and range?

Hint: write $x+2$ as $x-(-2)$. Draw the two asymptotes dashed first; they cross at (h, k). The points $(1, 1)$ and $(-1, -1)$ on $y = \frac{1}{x}$ move with them.



$h =$ _____ $k =$ _____

The *translation*: ____ units _____ and ____ units _____.

vertical asymptote: $x =$ _____

horizontal asymptote: $y =$ _____

domain: _____

range: _____

Two points on your graph: _____ _____

Practice: From a described translation to the equation

The graph of $y = \dfrac{1}{x}$ is translated 5 units down and 1 unit to the left.

a. In $y = \dfrac{1}{x-h} + k$, $h =$ _____ and $k =$ _____. A move to the *left* makes h _____.

b. Which is an equation of the translated graph? Circle one.

A. $y = \dfrac{1}{x+1} - 5$ B. $y = \dfrac{1}{x-1} - 5$ C. $y = \dfrac{1}{x-5} + 1$ D. $y = \dfrac{1}{x+5} - 1$

c. Its *vertical asymptote* is $x =$ _____ and its *horizontal asymptote* is $y =$ _____.

d. Why is the sign in choice B a trap?

_____

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
ULD ★	Domain and range of reciprocal functions	Quiz 4: asymptotes, domain and range of $y = \frac{1}{x-h} + k$ (Example 5)
F6J ★	Translations of functions	Quiz 5; Topic test Questions 5 and 14: a translation and its equation (Example 5, the Practice)
PNY ★	Write and solve inverse variation equations	Quiz 3 (bonus); Topic test Question 1: find k, then divide (the Extension)

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Extension (self-paced): Practice 20 (DOK 3), Model With Mathematics

The time t required to empty a water tank varies inversely as the rate of pumping p. A pump can empty a water tank in 40 min at the rate of 120 gal/min. Write the equation of the inverse variation. How long will it take the pump to empty the water tank at the rate of 200 gal/min? Hint: when t varies inversely as p, the product tp is always the same number k. Find k from the pair you know, then divide.

Equation: $t =$ _____ Time at 200 gal/min: _____ min

Before the quiz, finish these sentences.

The graph of $y = \dfrac{1}{x-h} + k$ is a *translation* of $y = \frac{1}{x}$: _____ units horizontally and _____ units vertically.

Its *vertical asymptote* is _____ and its *horizontal asymptote* is _____.

So its *domain* leaves out _____ and its *range* leaves out _____.

A plus sign inside the denominator, as in $x + 6$, means the graph moved to the _____, because _____.